

# What the Headlines Say

## Measuring Cognitive Bias and Agency Framing in AI News Discourse

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### 1 Introduction

Most people’s understanding of artificial intelligence comes not from building these systems or reading technical papers, but from the headlines they scroll past each morning. This matters because those headlines are not neutral. A headline reading “*AI Takes Over Another Industry*” frames reality very differently from “*Companies Adopt AI Automation Tools*,” even when both describe the exact same event. The first positions AI as the one in charge and acting. The second describes a deliberate human choice. Over millions of headlines and months of coverage, those small language differences accumulate into a public portrait of what AI is, who controls it, and what it means for human life.

This project analyzes 179,372 AI news headlines scraped from 13,019 publishers across 20 monthly windows from August 2024 to March 2026. Three original analytical frameworks were built to measure how these headlines are structured. The **Power Shift Index (PSI)** tracks whether AI or people are the ones acting and in control in a sentence. The **AI Anxiety Index (AAI)** measures how much fear and risk language a headline carries, using four weighted vocabulary categories drawn from psychology and media framing research. The **Invisible Human Framework (IH)** checks whether headlines engage with any of the things that make life actually human: grief, love, dignity, community, and more. Together, these three frameworks answer a simple question: what story is AI news actually telling?

The Stanford AI Index 2025 reports that global nervousness about AI products rose 13 percentage points between 2022 and 2024, while public confidence in AI companies to handle personal data fell from 50% to 47% [1, 2]. These shifts were not produced by direct experience with AI systems. For most people, they reflect what headlines say. That is why framing matters at scale.

From our analysis, **three findings** stand out. First, across all 20 months, AI appeared as the one in control roughly 7 times more often than a human did, though that gap is slowly narrowing. Second, headlines about AI safety and risk carry five times more fear language than headlines about AI products, but even product coverage is not anxiety-free. Third, and most strikingly, 92.4% of all 179,372 headlines contain zero words from any human-experience category, a rate that does not change even when the headlines are specifically about workers, researchers, or users.

#### 1.1 Research Questions

Three questions drive this project. First: is AI news coverage increasingly treating AI as the one in control, with humans fading into the background? Second: how concentrated is anxiety and risk language, and which parts of the news landscape carry the most of it? Third: when AI coverage mentions humans, does it engage with what makes human life actually human, grief, love, purpose, dignity, community, or does it treat people as inputs to a system? These questions are answered through PSI, AAI, and the Invisible Human Framework respectively.

#### 1.2 Course Connections

This project spans the full CS 210 pipeline: scraping with custom query logic and checkpoint-based logging, deduplication and loading into a normalized SQLite database that separates raw data from derived features, feature engineering through VADER sentiment analysis and sentence-transformer embeddings with three custom classifiers, predictive modeling with Ridge regression and a pruned decision tree, and multi-layered validation including gold-standard hand labeling, bootstrap confidence intervals at  $n = 179,372$ , 200-permutation hypothesis testing, and cluster stability testing across seeds, subsamples, and feature subsets.

### 1.3 Novel Contributions

Prior work on AI media framing is limited in scale and methodological consistency. Nguyen and Hekman [3] studied framing across only four outlets. Rodrigo-Gines et al. [4] and Castillo-Campos et al. [5] both concluded the field lacks reproducible quantitative metrics for the psychological texture of coverage. This project introduces PSI, AAI, and the Invisible Human Framework: three original, documented metrics, each with calibration history and multi-method validation, applied to a corpus two orders of magnitude larger than most prior work.

## 2 Data Collection and Processing

### 2.1 Scraping Architecture

Headlines were collected using the `gnews` Python library, which queries Google News RSS feeds without an API key, making the full collection reproducible from scratch. The scraper ran locally with checkpoint-based logging because the full run took 667 minutes (11.1 hours) and cloud sessions would drop mid-execution. Every completed monthly window was written immediately to a checkpoint CSV, so a failure partway through would not lose earlier progress.

Each of 192 keyword queries was run against each of 20 monthly windows, totaling 3,840 API calls with zero failures. All checkpoint files were merged and deduplicated on exact title strings after lowercasing and whitespace stripping. A cross-window pass removed 395 headlines that appeared in more than one month, keeping each in its earliest window.

Why monthly windows? The `gnews` API caps results at 100 per query call. A single broad date range returns only the most recent 100 headlines, making earlier content unreachable. Monthly windows force the API to surface content from each specific time period. This design extends the Figshare baseline approach [13] from 3 months to 20.

### 2.2 Query Architecture

192 search queries were organized into 9 thematic tiers. **The keywords were selected after reading psychology and media framing literature on how AI gets discussed in public discourse, ensuring the corpus covered not just product launches but also the emotional, political, and human dimensions of AI coverage.**

Table 1: Query architecture by tier

| Tier               | n  | Example queries                | Tier                  | n          | Example queries          |
|--------------------|----|--------------------------------|-----------------------|------------|--------------------------|
| T1: Core Terms     | 13 | artificial intelligence, LLM   | T6: Policy/Governance | 19         | EU AI Act, AI regulation |
| T2: Applications   | 57 | AI healthcare, AI education    | T7: Human Impact      | 19         | AI grief, AI loneliness  |
| T3: Risk and Fear  | 21 | AI safety, AI existential risk | T8: Democracy/Disinfo | 12         | AI deepfake, AI election |
| T4: Agency         | 17 | AI decides, AI takes over      | T9: Infrastructure    | 12         | AI energy, data centers  |
| T5: Named Entities | 25 | ChatGPT, DeepSeek, Gemini      | <b>Total</b>          | <b>192</b> |                          |

Tier 7 (Human Impact) is methodologically significant. Those 19 queries explicitly searched for headlines using the vocabulary of human experience. The fact that Tier 7 returned proportionally few unique headlines, and that most of those headlines still scored zero on the Invisible Human Framework, is one of the project’s central findings. The absence holds even when the search was actively looking for it.

## 2.3 Final Dataset

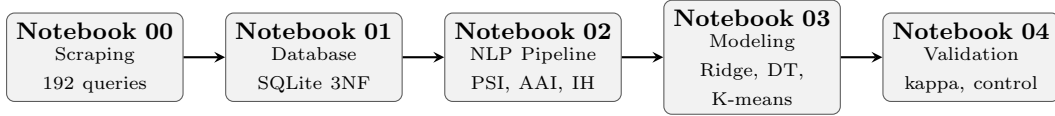
**Table 2:** Corpus overview

| Metric                | Value   | Metric                | Value                |
|-----------------------|---------|-----------------------|----------------------|
| Total headlines       | 179,372 | Date range            | Aug 2024 to Mar 2026 |
| Unique publishers     | 13,019  | Monthly windows       | 20                   |
| Defensible publishers | 593     | Largest monthly vol.  | 11,335 (Mar 2026)    |
| Max publisher share   | 2.4%    | Smallest monthly vol. | 6,500 (Aug 2024)     |

After all deduplication, the corpus grew 74% over the observation period, from 6,500 headlines in August 2024 to 11,335 in March 2026. January 2025 shows a notable spike of 7,942 headlines, coinciding with the DeepSeek release and the surge of coverage it triggered. Volume stayed above 10,000 per month from October 2025 onward, reflecting how central AI had become to the news cycle. Publisher breadth is reported as 593 outlets contributing 50 or more headlines rather than the raw 13,019, which includes single-mention aggregator entries that inflate the count.

## 3 Methodology

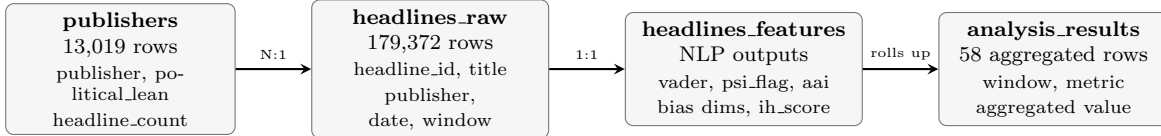
The project was organized across five notebooks, each with a single non-overlapping responsibility. No notebook modifies any earlier notebook’s outputs, making the pipeline fully reproducible at any stage.



**Figure 1:** Five-notebook pipeline. Each notebook writes only to its designated output table or file.

### 3.1 Database Schema

The data is stored in SQLite in third normal form across four tables, connected in a single left-to-right pipeline from raw input through computed features to aggregated output.



**Figure 2:** SQLite schema: a single left-to-right pipeline. Publishers seed the raw corpus. Each headline links 1:1 to its computed NLP scores. Features roll up into 58 monthly summary rows. **headlines\_raw** is immutable after scraping; all computed scores live in **headlines\_features**. Five indexes on high-traffic columns reduce monthly JOIN queries from 40+ seconds to under 2 seconds.

Separating raw data from derived features was a deliberate and necessary design choice. The PSI scoring formula went through three full calibration rounds. Because computed scores live in **headlines\_features** and never touch **headlines\_raw**, each revision only required dropping and recreating the features table, preserving the full corpus and avoiding hours of rescraping every time the methodology changed.

### 3.2 Sentiment Analysis: VADER

VADER [7] was selected for headline-level sentiment scoring after evaluating three alternatives. GPT-4o would cost approximately \$180 for 179,372 headlines and is not reproducible since the model updates silently between API releases. FinBERT [12] is fine-tuned on financial text and introduces domain bias when applied to safety and policy coverage. A local RoBERTa model requires 4–6 hours on CPU with minimal gains on short headlines. VADER is specifically validated for short informal text, runs in 25 seconds on the

full corpus, is free and deterministic, and its compound score ranges from  $-1$  (most negative) to  $+1$  (most positive) with direct interpretability.

On this corpus, VADER found the average headline leans slightly positive (mean compound  $+0.059$ ). Breaking that down: 36% of headlines carry predominantly positive language, 21.7% negative, and 42.2% neutral. That means AI coverage skews toward cautiously upbeat language overall, but with a significant negative minority, a pattern that becomes more interesting when compared against the anxiety and framing scores.

### 3.3 Cognitive Bias Detection

After personally reading through psychology and media framing literature [10, 11] on how narratives shape public perception of technology, eight cognitive bias dimensions were selected, each representing a distinct and detectable pattern in how language can frame AI. Understanding what each one means is important for interpreting the results:

- **Fear bias:** Coverage that frames AI as dangerous or threatening, often without proportionate evidence.
- **Optimism bias:** Uncritical celebration of AI capabilities with limited acknowledgment of downsides or uncertainty.
- **Anthropomorphism:** Describing AI as though it has human-like intent, feelings, or consciousness (e.g., “AI wants,” “AI fears,” “AI thinks”).
- **Moral panic:** Treating AI as an urgent, existential social threat requiring immediate collective action.
- **Agency attribution:** Crediting AI with autonomous decision-making or independent action (“AI chose,” “AI decided”).
- **Techno-utopianism:** The belief that AI will inevitably solve humanity’s biggest problems, disease, poverty, climate, without examining whether or how.
- **Economic threat:** Framing AI primarily through the lens of job loss and economic disruption.
- **Control loss:** Portraying humans as losing, or having already lost, meaningful control to AI systems.

Each dimension uses a hybrid scoring approach that combines two methods: (1) semantic similarity, comparing how closely a headline matches a prototype sentence written for that bias dimension using a sentence-transformer model (all-MiniLM-L6-v2 [8]), and (2) a keyword presence bonus weighted at 0.20 for direct vocabulary matches.

Three approaches were tried before settling on this hybrid. **Keyword-only scoring** was fast but brittle. For example, “*AI breaking records*” scored high on economic threat because “breaking” matched damage vocabulary regardless of what the headline was actually saying. Any word in the lexicon, even in a completely unrelated sense, triggered a flag. **Embedding-only scoring** was semantically richer but missed headlines that used clear bias vocabulary if the overall sentence structure did not closely resemble the prototype. The **hybrid** solves both problems at once. To check whether it actually worked better, all three methods were compared: keyword-only and embedding-only scores had almost no relationship to each other (Pearson correlation  $r = 0.07$ , meaning they were essentially measuring different things), while the hybrid and embedding-only correlated at  $r = 0.96$ , meaning the hybrid preserved semantic richness while the keyword bonus caught the obvious cases the embeddings missed.

A detection threshold of 0.50 was set through hands-on calibration: 100 headlines were scored and hand-reviewed to find the cutoff that minimized both false positives (flagging a bias that was not there) and false negatives (missing one that was). At 0.40, roughly 1 in 4 flagged headlines was a false alarm. At 0.50, that dropped to 1 in 8, with 87% agreement between the classifier and the hand labels. At 0.60, false alarms were rare but the classifier started missing real biases at a 19% rate. The 0.50 threshold was the best trade-off. A stress test confirmed the classifier’s stability: randomly removing 20% of the lexicon keywords 100 times in a row barely changed the scores, with correlations staying above  $r = 0.90$  across all eight dimensions.

### 3.4 Power Shift Index (PSI)

PSI asks a simple question about each headline: is AI portrayed as the one in control, or is it a person? Each headline is assigned one of four labels. **AI is in control** (PSI: AI\_AGENT) means an AI-related word appears close to an action verb, so AI is the one acting. **Person is in control** (PSI: HUMAN\_AGENT) means a human word precedes the verb, so a person is the one doing. **Both** (PSI: DUAL\_AGENT) means both appear. **Neither** (PSI: NEUTRAL) means no clear actor was detected. Examples from the corpus:

- *AI in control*: “2025 Mass Layoffs: AI, Automation, and Restructuring Hit Global Jobs Market” (CXO Digitalpulse)
- *Person in control*: “Chinese researchers used Llama 13B to build chatbot for military use” (SiliconANGLE)
- *Both*: “OpenAI CEO Sam Altman Warns Users Against Blind Trust in AI” (Elets CIO)

$$\text{PSI} = \frac{\text{count}(\text{AI in control})}{\max(\text{count}(\text{Person in control}), 1)} \times 100 \quad (1)$$

**A PSI of 100 means balanced coverage: equal numbers of AI-driven and human-driven headlines. Any score above 100 means AI is being portrayed as in control more often than people are. The overall corpus PSI is 733.**

Building the classifier required three attempts. **Version 1** recognized only five AI subject tokens (AI, model, system, algorithm, robot), which made ChatGPT, Gemini, DeepSeek, and every named AI system invisible. About 27% of genuine AI-in-control headlines were missed entirely. **Version 2** expanded the token list but used a verb list so broad that compound adjectives like “AI-powered” inflated counts incorrectly. **Version 3** fixed both: negation detection to avoid flagging “AI cannot decide,” a stricter verb list, an exclusions list for compound adjectives, and a separate Both label for co-occurrence cases. Final distribution: 8,620 AI in control (4.8%), 1,176 Person in control (0.7%), 189 Both (0.1%), 169,387 Neutral (94.4%).

### 3.5 AI Anxiety Index (AAI)

AAI was built from scratch to measure how much fear, risk, and worry language a headline carries. A manual review of 500 randomly sampled headlines identified four vocabulary categories based on how directly and clearly each one signals anxiety in AI news:

$$\text{AAI}_{\text{raw}} = 1.0 \cdot V_{\text{risk}} + 0.8 \cdot V_{\text{health}} + 0.6 \cdot V_{\text{uncertain}} + 1.2 \cdot V_{\text{fear}} \quad (2)$$

- **Risk vocabulary** ( $w=1.0$ ): *risk, threat, danger, hazard, peril, unsafe*. Direct threat language and the reference baseline.
- **Health and safety** ( $w=0.8$ ): *health, medical, injury, lethal, safety*. Weighted slightly below risk to avoid double-counting headlines that contain both “danger” and “safety.”
- **Future uncertainty** ( $w=0.6$ ): *could, might, may, potential, uncertain*. Given the lowest weight because these words appear just as often in neutral headlines as anxious ones, so they are a weak signal on their own.
- **Fear intensifiers** ( $w=1.2$ ): *terrifying, catastrophic, existential, panic, crisis*. Given the highest weight because when these words appear in an AI headline, the story is almost always about something alarming.

Raw AAI is capped at 3.6 (the maximum if all four categories fire) and normalized to  $[0, 1]$ :  $\text{AAI} = \text{AAI}_{\text{raw}}/3.6$ . Three headlines that score a perfect 1.0 illustrate why the weighting works: “*AI Chatbots Risk Mental Health Crisis, Stanford Study Warns*”, “*AI’s existential threat: Researcher predicts 99% unemployment by 2030 without safety measures*”, and “*Top AI safety expert warns that an unregulated AI arms race may pose existential risks*.” All four vocabulary categories fire in each of these headlines simultaneously.

### 3.6 Invisible Human Framework

The Invisible Human Framework asks a pointed question: AI is a technology built to affect human lives, so why does so little AI coverage use language that reflects what human lives actually feel like? Seven domains of lived human experience were defined with representative vocabulary. **Grief and Loss** (*grief, mourning, bereavement, funeral, widow*). **Spirituality and Meaning** (*spiritual, faith, soul, sacred, transcendence*). **Love and Relationships** (*love, romance, friendship, loneliness, intimacy*). **Bodily Experience** (*pain, pleasure, embodied, touch, disability*). **Dignity and Purpose** (*dignity, identity, self-worth, fulfillment, calling*). **Community and Belonging** (*community, solidarity, togetherness, collective*). **Childhood and Play** (*childhood, children, imagination, wonder, innocence*).

Each domain is flagged as either present or absent in a given headline. A key design distinction separates unconditional absence (checking all 179,372 headlines regardless of content) from conditional absence (check-

ing only the 29,896 headlines that explicitly mention workers, researchers, or users). The conditional result is the stronger claim: even when AI coverage is specifically about people, it still does not use the vocabulary of what those people actually experience.

### 3.7 Gold-Standard Validation

To check whether the automated classifiers actually captured what they were supposed to capture, **I hand-labeled 200 headlines myself**. These were drawn as a stratified sample, proportionally representing each PSI class and dominant bias category. Every headline was labeled one at a time, independently, without looking at what the model had predicted. That independence matters: if I had known the model’s answer, I might unconsciously have been influenced by it.

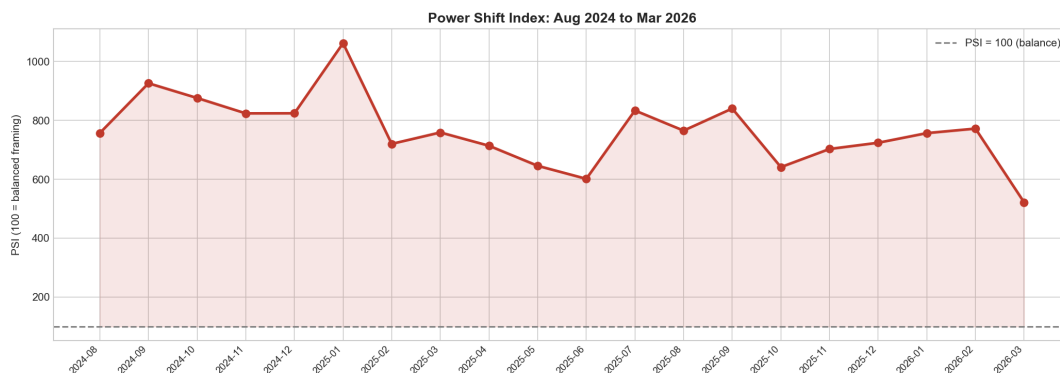
The level of agreement between my hand labels and the classifier is measured using Cohen’s kappa, a standard metric that accounts for the fact that some agreement happens by chance. A kappa of 0 means the classifier is no better than random guessing. A kappa of 1.0 means perfect agreement. A kappa around 0.4 is considered “fair,” meaning the classifier is doing meaningfully better than chance but is not highly precise. For PSI, kappa = 0.368, which falls in the fair range. Looking at the specific numbers: of 77 headlines I labeled as Person-in-control, the classifier only correctly identified 24 of them (31%), tagging the other 49 as Neutral. It did better on AI-in-control headlines, catching 47% of them. This means the PSI scores are conservative lower bounds: the true imbalance is likely larger than measured because person-in-control headlines are being undercounted.

For the eight cognitive bias dimensions, the overall agreement score (macro F1, which averages accuracy across all eight categories) was 0.706 across 134 labeled headlines. F1 ranges from 0 to 1, where 1 is perfect. Anthropomorphism was the easiest to classify correctly (F1 = 1.00), likely because phrases like “AI thinks” or “AI feels” are unambiguous. Techno-utopianism (F1 = 0.59) and optimism bias (F1 = 0.60) were the hardest, because utopian enthusiasm and general optimism look very similar in a short headline and are difficult to distinguish even by hand.

## 4 Results

### 4.1 Finding 1: AI is the one in control in most headlines, but the gap is slowly closing

**Across 20 months of coverage, for every headline where a person was driving the action, there were about seven where AI was.** That 7:1 ratio held throughout the entire observation period. The PSI score, which measures this imbalance (with 100 being perfectly equal), ranged from a low of 522 in March 2026 to a peak of 1,062 in January 2025 during the DeepSeek release, when coverage was dominated by stories about what DeepSeek was doing and capable of. Over time, the score showed a statistically significant downward trend of about 10.6 points per month, meaning the gap is narrowing.



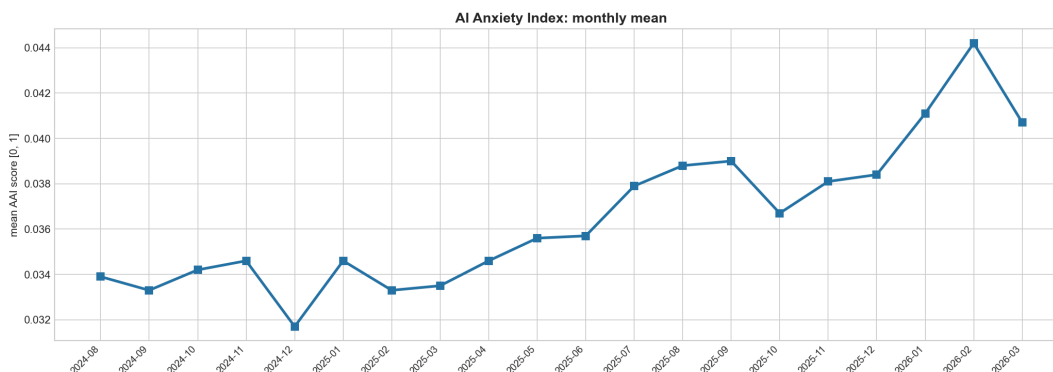
**Figure 3:** Power Shift Index by month, August 2024 to March 2026. The dotted line at 100 marks equal framing. Every monthly value is well above it, meaning AI is portrayed as in control far more often than people are.

That downward trend is not because individual AI stories started describing AI differently. It is because a whole new category of coverage grew rapidly: governance, regulation, and policy news. These stories are

structurally about what lawmakers, executives, and regulators are doing about AI, so they naturally place humans in the acting role. The framing of AI-capability stories did not change; it was diluted by a growing wave of stories where humans are the decision-makers.

#### 4.2 Finding 2: Safety headlines carry 5x the anxiety of product coverage, but even product news is not calm

The AI Anxiety Index score for the average headline is 0.037 on a 0 to 1 scale. That number seems small, and it is: 84.5% of all headlines score exactly zero because they contain none of the anxiety vocabulary. The average is pulled up entirely by a concentrated minority of headlines that score high. Importantly, that average has been rising steadily over the 20-month period, from 0.034 in August 2024 to 0.041 in March 2026, a reliable upward trend ( $R^2 = 0.77$ , meaning the trend accounts for 77% of the month-to-month variation in anxiety scores).



**Figure 4:** Mean AI Anxiety Index by month, August 2024 to March 2026. The steady rise ( $R^2 = 0.77$ ) means AI coverage has been getting consistently more anxious over time.

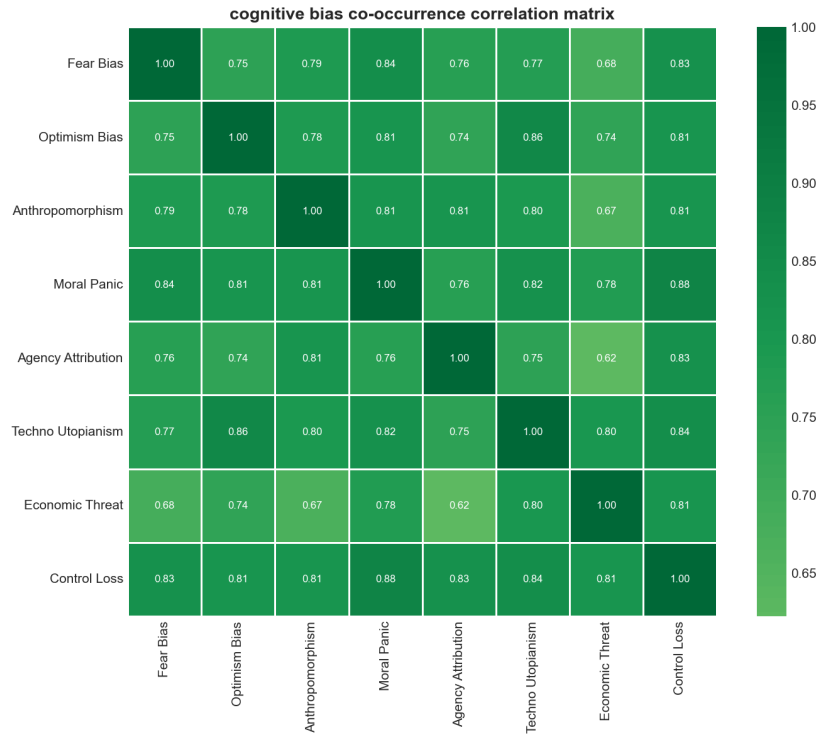
The gap between topic categories is striking. Safety and Risk headlines average five times the anxiety of product coverage (AAI 0.093 vs. 0.018); Health and Science headlines reach six times it (0.118). But the more meaningful observation is about the product headlines themselves: even routine coverage of new tools, partnerships, and earnings reports carries some residual fear vocabulary. The news ecosystem has developed a habit of treating AI as inherently risky, regardless of what the story is actually about.

**Table 3:** Mean AAI by topic category (selected). Higher scores mean more fear and risk language per headline.

| Category               | Headlines | Mean AAI |
|------------------------|-----------|----------|
| Health & Science       | 8,804     | 0.118    |
| Safety & Risk          | 15,793    | 0.093    |
| Society & Human Impact | 9,840     | 0.042    |
| Defense & Security     | 8,375     | 0.035    |
| Policy & Governance    | 12,128    | 0.027    |
| Companies & Products   | 28,155    | 0.018    |
| Business               | 10,617    | 0.017    |
| Creative Industries    | 10,814    | 0.017    |

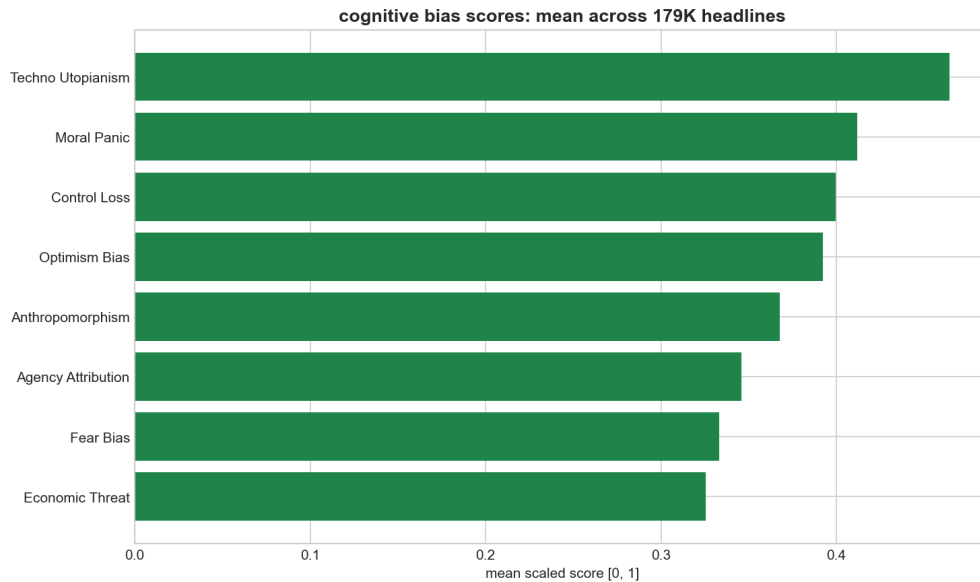
#### 4.3 Finding 3: AI headlines live in one of two worlds, breakthrough or crisis, with almost nothing in between

Each headline was scored across eight cognitive bias dimensions. When those scores are compared against each other, a clear pattern emerges: the negative-framing dimensions (fear bias, moral panic, control loss, and economic threat) are strongly correlated with each other, meaning headlines that sound fearful tend to sound panicked, threatened, and out-of-control all at once. The positive-framing dimensions (optimism bias and techno-utopianism) form an equally tight cluster on the other side. A headline tends to live entirely in one camp or the other. These two worlds rarely overlap.



**Figure 5:** How the eight bias dimensions relate to each other. Darker cells mean stronger correlation. The negative cluster (upper-left) and the positive cluster (lower-right diagonal of optimism and techno-utopianism) are both internally coherent and largely separate from each other.

Looking at the raw numbers, the most common framing across all headlines is techno-utopianism, the belief that AI will solve everything, which is the dominant tone in 64.7% of all headlines. Moral panic (AI as an urgent societal threat) is the dominant tone in 12.3%. What is almost entirely absent is the middle: nuanced coverage that acknowledges both potential and risk, that weighs trade-offs, or that describes AI in ordinary, context-specific terms.



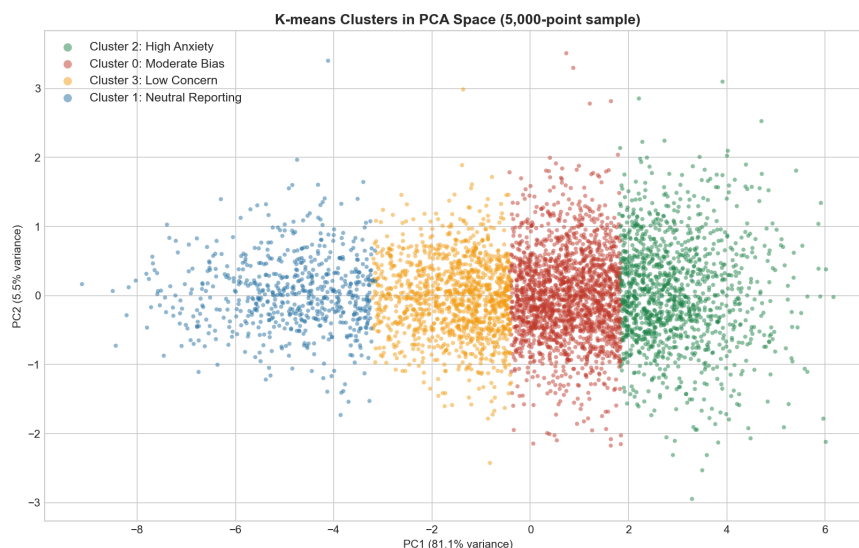
**Figure 6:** Average cognitive bias score per dimension across all 179,372 headlines. The fact that both techno-utopianism (utopian) and moral panic plus control loss (fearful) all score above the midpoint simultaneously confirms that AI coverage is not just one-sided: it is pulled hard in both directions at once.

#### 4.4 Finding 4: Only 1 in 8 AI headlines is genuinely neutral

To get a cleaner view of what kinds of stories AI journalism produces, K-means clustering was used to group the 179,372 headlines into natural categories based on their bias dimension scores. K-means works by finding groupings that are as internally consistent as possible. The goal is to identify whether headlines naturally fall into a few distinct types or exist on a smooth continuum. After testing cluster counts from 2 to 9, the data clearly organized into four groups. The groupings were tested for stability by running the algorithm repeatedly with different starting points and different subsets of the data, and the results were consistent every time (stability score of 0.941 out of 1.0).

**Table 4:** The four natural headline archetypes from clustering. Mean AAI shows how much anxiety language each type carries on average.

| Archetype         | Headlines | Share | Mean AAI |
|-------------------|-----------|-------|----------|
| High Anxiety      | 43,731    | 24.4% | 0.052    |
| Moderate Bias     | 68,132    | 38.0% | 0.038    |
| Low Concern       | 45,176    | 25.2% | 0.027    |
| Neutral Reporting | 22,333    | 12.5% | 0.025    |

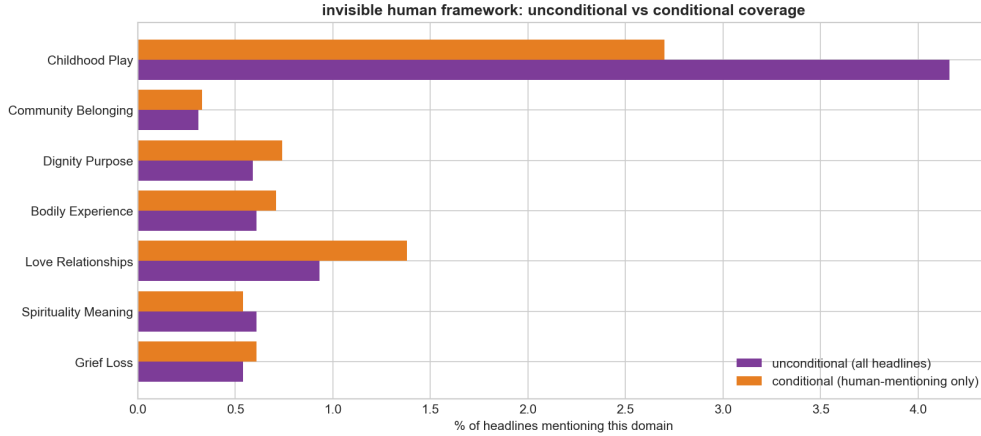


**Figure 7:** The four clusters visualized. Each dot is a headline, colored by cluster. The horizontal axis captures 81% of what makes headlines different from each other, and it cleanly separates the anxious headlines (left) from the neutral ones (right).

One in four AI headlines (24.4%) belongs to the High Anxiety group. That is not a fringe category: it is a structural quarter of everything published about AI. Add the Moderate Bias group (38%) and more than 60% of all AI headlines carry clear framing in one direction or another before a reader even opens the story.

#### 4.5 Finding 5: 92.4% of AI headlines contain zero words from any human-experience category

This is the central finding of the project. Out of 179,372 headlines about a technology that touches healthcare, work, education, relationships, and nearly every dimension of human life, 92.4% contain zero words from any of the seven human-experience categories. A headline can mention “workers” without using words like dignity, grief, or community. A headline can be about AI in elder care without mentioning pain, loss, or purpose. The vocabulary of what it means to be human is almost entirely absent from coverage of a technology reshaping what it means to be human.



**Figure 8:** How often each human-experience domain appears in headlines. The orange bars show all headlines; the purple bars show only headlines explicitly about workers, researchers, or users. The near-identical results confirm the absence is not incidental: even coverage about people does not use the language of human experience.

**Table 5:** Human-experience domain appearance rates across the full corpus

| Domain                 | Headlines | Rate  | 95% CI       |
|------------------------|-----------|-------|--------------|
| Childhood & Play       | 7,456     | 4.16% | [4.07, 4.25] |
| Love & Relationships   | 1,667     | 0.93% | [0.89, 0.97] |
| Spirituality & Meaning | 1,091     | 0.61% | [0.57, 0.64] |
| Bodily Experience      | 1,086     | 0.61% | [0.57, 0.64] |
| Dignity & Purpose      | 1,052     | 0.59% | [0.55, 0.62] |
| Grief & Loss           | 971       | 0.54% | [0.51, 0.58] |
| Community & Belonging  | 560       | 0.31% | [0.29, 0.34] |
| Any domain present     | 13,547    | 7.55% | [7.43, 7.67] |
| No domain present      | 165,825   | 92.4% | [92.3, 92.6] |

Childhood and Play is the most commonly appearing domain (4.16%), primarily because AI-in-education coverage uses words like “children” and “learning” regularly. Every other domain sits below 1%. Grief and Loss appears in just 0.54% of headlines about a technology actively deployed in elder care, end-of-life decisions, and bereavement support. The pattern is stable across all 20 months. No single event or news cycle drives it. It is structural.

A comparison corpus of 4,106 climate-change headlines, collected with the identical method, showed a 4.9% any-domain rate, versus AI’s 7.6%. Short headlines across all topic areas use less human-experience language than full articles do, so AI journalism is not uniquely cold relative to other technical beats. What is notable is that the absence holds even after specifically filtering for headlines about human beings.

Of the most anxious 10% of headlines, 90.9% still contain zero human-experience vocabulary. When AI coverage is most worried, it is also most stripped of the language of human consequence.

#### 4.6 Finding 6: What modeling revealed about data honesty

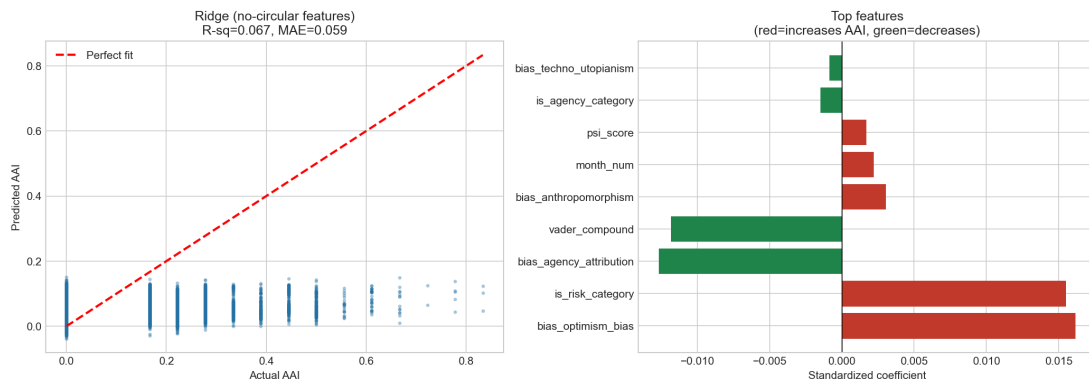
The modeling work on this project was as much about catching what the numbers could not honestly tell as about what they could.

**Predicting anxiety levels.** The first regression model using all 14 available features returned an  $R^2$  of 0.220, meaning the model could explain 22% of the variation in anxiety scores. That looked reasonable until closer inspection revealed a circularity problem: five of the 14 features (fear bias, moral panic, economic threat, control loss, and overall bias intensity) were derived from the same vocabulary lexicons as AAI itself. They were not independent predictors; they were partially reconstructing the answer from its own components. Removing those five features dropped  $R^2$  to 0.067. That 229% inflation was entirely an artifact of circular construction, not real predictive power.

The honest  $R^2$  of 0.067 means the model can explain about 7% of headline-to-headline variation in anxiety scores from genuinely independent information. A permutation test confirmed that this 7% is real: shuffling the anxiety scores randomly 200 times never produced a model that good by chance (empirical  $p < 0.005$ ). The features that genuinely predict higher anxiety are headlines about risk topics and those framed with utopian language, which co-occur with alarm vocabulary. Headlines with positive sentiment and those attributing specific actions to named humans show lower anxiety scores.

**Table 6:** Regression model comparison.  $R^2$  measures how much of the variation in anxiety scores each model explains. CV  $R^2$  is the cross-validated version, which checks whether the number holds on data the model did not train on.

| Model                          | $R^2$ | MAE   | CV $R^2$          |
|--------------------------------|-------|-------|-------------------|
| Ridge ( $\alpha = 1$ , honest) | 0.067 | 0.059 | $0.061 \pm 0.008$ |
| Lasso ( $\alpha = 0.01$ )      | 0.028 | 0.062 | $0.025 \pm 0.002$ |
| Random Forest                  | 0.153 | 0.054 | $0.140 \pm 0.013$ |
| Gradient Boosting              | 0.148 | 0.055 | $0.138 \pm 0.013$ |
| Ridge (with circular features) | 0.220 | 0.054 | —                 |



**Figure 9:** Left: how well the honest Ridge model predicts anxiety scores (each dot is a headline). Right: which features push anxiety up (red) or down (green). Risk topic and utopian framing increase predicted anxiety; positive sentiment and specific human attribution decrease it.

**Classifying headline bias type.** Five classifiers were tested on the task of predicting which of the eight bias categories dominates a given headline. Guessing randomly would get 12.5% right (1 in 8).

**Table 7:** Classifier comparison. Train and Test columns show accuracy on training data vs. unseen data. A large gap between them means the model memorized the training data rather than learning a generalizable pattern (overfitting).

| Model                             | Train | Test  | Gap   |
|-----------------------------------|-------|-------|-------|
| Logistic Regression               | 0.996 | 0.995 | 0.001 |
| Decision tree (unpruned)          | 1.000 | 0.971 | 0.029 |
| Random Forest                     | 0.838 | 0.831 | 0.007 |
| KNN ( $k = 15$ )                  | 0.881 | 0.864 | 0.017 |
| Decision tree (pruned, depth = 5) | 0.745 | 0.744 | 0.001 |

Two models looked suspiciously good. Logistic Regression at 99.5% accuracy is misleading: it predicts the bias class from embedding-derived features that were used to generate the bias labels in the first place, creating a near-closed loop. The unpruned decision tree at 100% training accuracy is a textbook case of overfitting, where the model memorized the training data rather than learned a real pattern. Its 2.9 percentage point drop on unseen data confirms it.

The pruned decision tree (depth limited to 5 levels) was the right choice. Limiting depth prevents memorization structurally, without needing manual tuning. It reaches 74.4% accuracy on unseen data, nearly six times better than random guessing, and the near-zero gap between training and test accuracy confirms it is learning something real. Its decisions are also human-readable, which matters for understanding what the classifier actually learned. The main limitation is class imbalance: because techno-utopianism dominates 64.7% of all headlines, the classifier learned that category well ( $F1 = 0.81$ ) and struggled with the rare categories like economic threat and agency attribution.

#### 4.7 Cross-Metric Analysis

PSI, AAI, and the Invisible Human Framework each measure something genuinely different. PSI and AAI have almost no correlation ( $r = +0.013$ ) across the full corpus: a headline can portray AI as in control without being anxious, and vice versa. **None of the three metrics is redundant with the others.**

The most striking cross-metric finding: among the most anxious 10% of headlines, 90.9% still contain zero human-experience vocabulary. When coverage is most worried about AI, it is also most stripped of any language connecting those worries to the human lives at stake. Publisher concentration does not drive any of these results. The top 10 publishers and the long tail of smaller outlets show nearly identical AAI scores (0.040 vs. 0.037), confirming these patterns are not artifacts of a few dominant voices.

### 5 Hypothesis Tests

Five hypotheses were written down before running any inferential tests. Two were supported; three were not, and what each non-result reveals is informative.

**Table 8:** Pre-registered hypothesis test results

| Hypothesis                                             | Verdict             | Effect       | p-value       |
|--------------------------------------------------------|---------------------|--------------|---------------|
| H1: PSI shows upward trend                             | Reversed (declined) | $R^2 = 0.28$ | 0.017         |
| H2: Risk carries higher anxiety than Products          | Supported           | $d = 0.749$  | $\approx 0$   |
| H3: Named entities carry lower anxiety than generic AI | Supported           | $d = -0.133$ | $< 10^{-130}$ |
| H4: Human-mentioning headlines use more IH vocabulary  | Null                | $d = -0.032$ | $< 0.001$     |
| H5: Model release months have higher PSI               | Underpowered        | $d = 0.852$  | 0.713         |

**H1** predicted PSI would rise over time, since AI capabilities were expanding and stories about what AI was doing would multiply. Instead, PSI declined significantly. The reason is that a separate category of coverage, governance, regulation, and policy stories, grew even faster and placed human actors at the center of its sentences. The surprise reversal is itself a finding: AI policy coverage is growing faster than AI capability coverage.

**H2** is confirmed with a meaningful effect size. Safety and risk headlines carry five times more anxiety vocabulary than product and business headlines on average. The difference is large enough to matter in practice, not just in a statistical sense.

**H3** is confirmed with a smaller but still highly significant effect. Headlines naming a specific AI product (ChatGPT, Gemini, Claude) use less generalized anxiety language than headlines that say only “AI.” Giving the technology a specific name and context grounds the story in something concrete and reduces the ambient threat framing that comes with generic language.

**H4** was a check on internal consistency: headlines that mention people (workers, researchers, users) should use more human-experience vocabulary than headlines that do not. The effect was negligible and pointed slightly in the wrong direction. This matters because it shows that mentioning a person functionally (“the worker,” “the user”) is completely different from describing that person experientially (“grief,” “dignity,” “belonging”). AI coverage writes about people without writing about people as people.

**H5** predicted that months with major model releases would show higher PSI, since those releases dominate coverage with AI-as-actor framing. The effect size was large ( $d = 0.852$ ) but the result was not statistically significant ( $p = 0.713$ ) because only two clear release months fell within the 20-month window. This is an underpowered result, not evidence against the hypothesis.

## 6 Limitations

**PSI classifier precision.** The fair kappa (0.368) means person-in-control headlines are systematically undercounted: about a quarter of them are tagged as Neutral. All PSI values are conservative lower bounds; the true framing imbalance is likely larger. The trend findings remain valid because the same imprecise classifier was applied identically across all 20 months.

**Invisible Human ambiguity.** Mentioning a person functionally and describing a person experientially turned out to pick out nearly disjoint sets of headlines (H4). The 92.4% absence is a well-supported descriptive claim about what words appear in headlines, not a claim about journalistic intent. The climate comparison confirms that short headlines across all genres use limited human-experience vocabulary.

**Publisher count.** The raw 13,019 unique publishers includes single-mention aggregator entries. All publisher breadth claims use the 593 outlets with 50 or more headlines.

**Decision tree class imbalance.** The 74.4% accuracy is substantially driven by the techno-utopianism majority. Rare bias categories (economic threat, agency attribution) are near-undetectable at current class distributions and would require targeted oversampling to improve.

**Political lean sub-sample.** AllSides matched 118 of 13,019 publishers. All correlations between political lean and PSI/AAI/IH were negligible ( $|\rho| < 0.05$ ). No claims about political asymmetry in framing are made.

## 7 Conclusion

This project set out to measure what story AI news is actually telling at scale. Across 179,372 headlines from 13,019 publishers over 20 months, three consistent patterns emerged.

**AI is portrayed as the one in control.** For every headline where a person was driving the action, about seven showed AI doing it. Headlines like “*AI predicts first round picks*” (USA Today) or “*AI restructuring hits global jobs market*” position the technology as the actor causing things to happen. Headlines where a person is clearly in charge, like “*Chinese researchers used Llama 13B to build a military chatbot*,” are far less common. Repeated exposure to AI-as-actor framing normalizes the idea that AI operates independently of human decision-making, and that gap is real regardless of whether it reflects technical reality.

**AI news is getting more anxious, and anxiety is concentrated but not contained.** Headlines about AI safety carry five times more fear vocabulary than product headlines, but even routine product coverage is not anxiety-free. The news ecosystem has developed a habit of treating AI as inherently risky regardless of context. Only 1 in 8 headlines qualifies as genuinely neutral: 87.5% of all coverage carries detectable framing toward either utopian promise or existential concern, with very little in between.

**Human experience is almost entirely absent.** 92.4% of all headlines contain zero words from any domain of lived human experience. A headline like “*AI’s existential threat: Researcher predicts 99% unemployment by 2030*” raises alarm without asking what that would mean for people’s sense of purpose, community, or identity. The most anxious headlines are also the most abstract. When coverage is most worried about what AI might do, it is simultaneously most silent about what it would actually mean for a person to live through it.

The modeling phase reinforced the importance of honest accounting. A regression model initially appeared to explain 22% of anxiety-score variation, but five features turned out to be circularly derived from the same data as the target. Removing them dropped the honest figure to 7%, confirmed as real signal by a 200-permutation test. The broader lesson: a high model score can be a sign of circular construction rather than genuine predictive power, and checking for it is as important as reporting the final result.

Looking back, the most surprising result was not the 92.4% Invisible Human rate, which is striking, but the H4 null finding that confirmed it in a different way. The intuition going in was that headlines explicitly

mentioning workers or researchers would score higher on human-experience vocabulary than ones that did not. They did not. Mentioning a person functionally, as a worker or a user, turns out to be completely different from describing that person as a human being with inner life. That distinction, between covering people and humanizing people, is one that no prior metric in the AI journalism literature had put a number on. It is also the finding that has the clearest implication for anyone writing about AI: adding the word “worker” to a headline does not mean you have written about a person.

## 7.1 Future Work

**A better PSI classifier.** The current PSI classifier uses a token-proximity approach: it looks for AI-related words within six tokens of an action verb. This works for simple active-voice headlines but misses passive constructions entirely. A headline like “Workers were displaced by AI-driven automation” puts humans at the front of the sentence but attributes the action to AI further in, which the current classifier would tag as Neutral. A dependency-parser-based approach, one that follows grammatical relationships rather than word proximity, would correctly classify these cases. Given that the hand-labeled kappa of 0.368 suggests roughly a quarter of person-in-control headlines are being missed, a better classifier would likely reveal a framing imbalance larger than the 7:1 reported here.

**Supervised classification for the Invisible Human Framework.** The current IH framework uses keyword matching: a headline is flagged for a domain if any of its vocabulary words appear. That approach is precise but narrow. A headline that describes grief without using the word grief, such as “*AI takes the last job a father had to show his son*”, would score zero under the current framework. Supervised classification, trained on a gold-standard labeled set of 300 to 500 headlines per domain, could learn the broader semantic patterns of human-experience language and would almost certainly find higher domain presence rates than keyword matching does. Given that keyword matching is a lower bound by construction, the 92.4% absence rate is a conservative figure.

**Cross-medium and cross-language extension.** This corpus covers English-language print news scraped from Google News. That is a significant portion of the public information environment, but it excludes television, podcasts, social media, non-English press, and the wire services (AP, Reuters) whose framing gets picked up and amplified by thousands of outlets. A cross-medium study would test whether the same PSI and AAI patterns hold in broadcast formats, where conversational language and interview structure might pull framing differently. A cross-language extension would test whether the bifurcated framing pattern (breakthrough or crisis) is a feature of English-language journalism specifically or of AI coverage globally.

**Framing over longer time horizons.** The 20-month window in this corpus captured the period from the initial post-ChatGPT AI boom through the beginning of AI regulation. The declining PSI trend is promising but covers too short a window to be confident it represents a structural shift rather than a temporary compositional effect. A follow-up collection in 2027 or 2028 would test whether PSI continues declining as regulation matures, whether AAI keeps rising as governance debates intensify, and whether the Invisible Human domains remain near zero even as conversations about AI’s social consequences become more prominent.

## Project Resources

- **Interactive Website:** [https://sadhanavasanthakumar.github.io/sv903\\_FinalProject/](https://sadhanavasanthakumar.github.io/sv903_FinalProject/)
- **GitHub Repository:** [https://github.com/SadhanaVasanthakumar/sv903\\_FinalProject/](https://github.com/SadhanaVasanthakumar/sv903_FinalProject/)
- **Project README:** [https://github.com/SadhanaVasanthakumar/sv903\\_FinalProject/blob/main/README.md](https://github.com/SadhanaVasanthakumar/sv903_FinalProject/blob/main/README.md)
- **Full Appendix:** [https://github.com/SadhanaVasanthakumar/sv903\\_FinalProject/blob/main/APPENDIX.md](https://github.com/SadhanaVasanthakumar/sv903_FinalProject/blob/main/APPENDIX.md)
- **Demo Video:** [Link to be added]

**AI Tool Disclosure.** cGPT was used for coding syntax questions and to help build the interactive website. All research questions, analytical frameworks, metric definitions, lexicon construction, hypothesis formulation, statistical analysis, and written content are entirely my own work.

**Additional Figures.** All 19 generated figures are available in the website gallery under the “All Figures” tab and in `outputs/` in the GitHub repository. Figures not included due to space: decision tree confusion matrix, feature importance chart, Invisible Human temporal stability chart, K-means elbow and silhouette plots, bias co-occurrence network graphs, and PSI/AAI extrapolation panels.

## References

- [1] Stanford HAI. *AI Index Report 2024*. Stanford University, 2024.
- [2] Stanford HAI. *AI Index Report 2025*. Stanford University, 2025.
- [3] Nguyen, D., and Hekman, E. The news framing of artificial intelligence: A critical exploration of how media discourses make sense of automation. *AI and Society*, 39(2), 437–451, 2024.
- [4] Rodrigo-Gines, F.-J., Carrillo-de-Albornoz, J., and Plaza, L. A systematic review on media bias detection. *Expert Systems with Applications*, 237, 121641, 2024.
- [5] Castillo-Campos, M., Becerra-Alonso, D., and Boomgaarden, H. Automated detection of media bias using AI and NLP: A systematic review. *Social Science Computer Review*, 2025.
- [6] Darling, K. “Who’s Johnny?” Anthropomorphic framing in human-robot interaction, integration, and policy. In *Robot Ethics 2.0*. Oxford University Press, 2017.
- [7] Hutto, C. J., and Gilbert, E. VADER: A parsimonious rule-based model for sentiment analysis of social media text. *AAAI ICWSM*, 2014.
- [8] Reimers, N., and Gurevych, I. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *EMNLP*, 2019.
- [9] Landis, J. R., and Koch, G. G. The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159–174, 1977.
- [10] Cave, S., and Dihal, K. Hopes and fears for intelligent machines in fiction and reality. *Nature Machine Intelligence*, 1(2), 74–78, 2019.
- [11] Brennen, J. S., Howard, P. N., and Nielsen, R. K. *An Industry-Led Debate: How UK Media Cover Artificial Intelligence*. Reuters Institute for the Study of Journalism, 2018.
- [12] Araci, D. FinBERT: Financial sentiment analysis with pre-trained language models. *arXiv:1908.10063*, 2019.
- [13] AI News Headlines Dataset. Figshare, 2024. [https://figshare.com/articles/dataset/Dataset\\_of\\_AI-related\\_News\\_Headlines\\_Aug\\_2024\\_Jul\\_2025\\_/30189736](https://figshare.com/articles/dataset/Dataset_of_AI-related_News_Headlines_Aug_2024_Jul_2025_/30189736). Used as methodological baseline; this project extends it from 89 days and 20,581 headlines to 607 days and 179,372 headlines.